

CAD Project

PWM Generator

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1. Project specifications

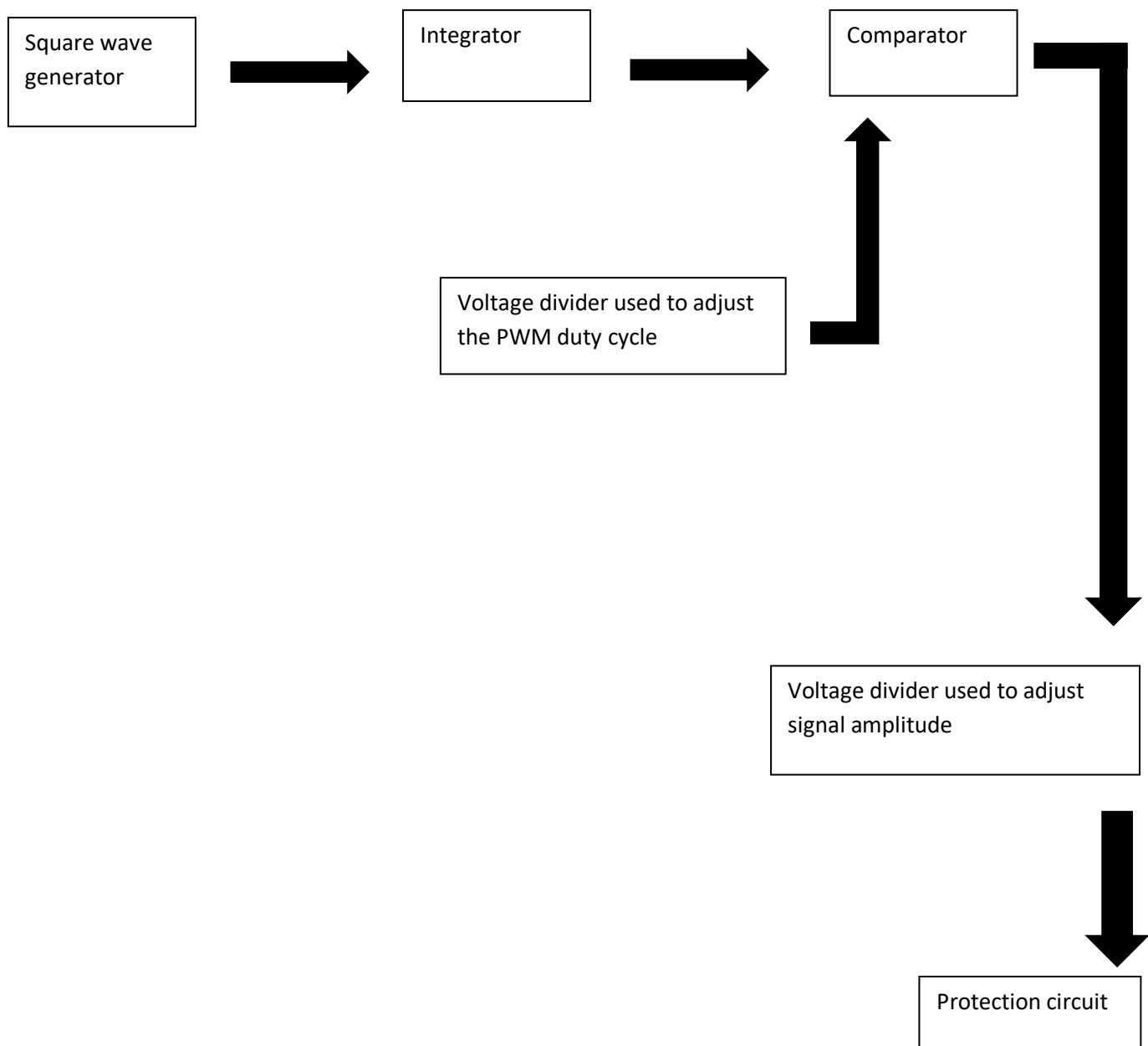
The presented circuit is a PWM (Pulse Width Modulation) generator, designed to deliver a digital signal with controllable duty cycle and frequency. It is composed of several key functional blocks:

- **Square wave generator** – produces a stable square wave signal that serves as the basis for further waveform shaping.
- **Integrator** – converts the square wave into a triangular waveform, essential for accurate PWM generation.
- **Comparator** – compares the triangular waveform with a reference voltage to generate the PWM signal.
- **Duty cycle adjustment voltage divider** – allows fine-tuning of the reference voltage, thereby enabling control over the PWM signal's duty cycle.
- **Amplitude adjustment voltage divider** – regulates the output signal's amplitude to match specific application requirements.
- **Protection circuit** – safeguards the load and sensitive components by preventing overcurrent conditions and voltage anomalies.

Values for the circuit are:

fu1	30%
fu2	60%
APWM	4
APWM2	5
f[Hz]	2000

2. Block diagram



3. The electrical schematic of the circuit

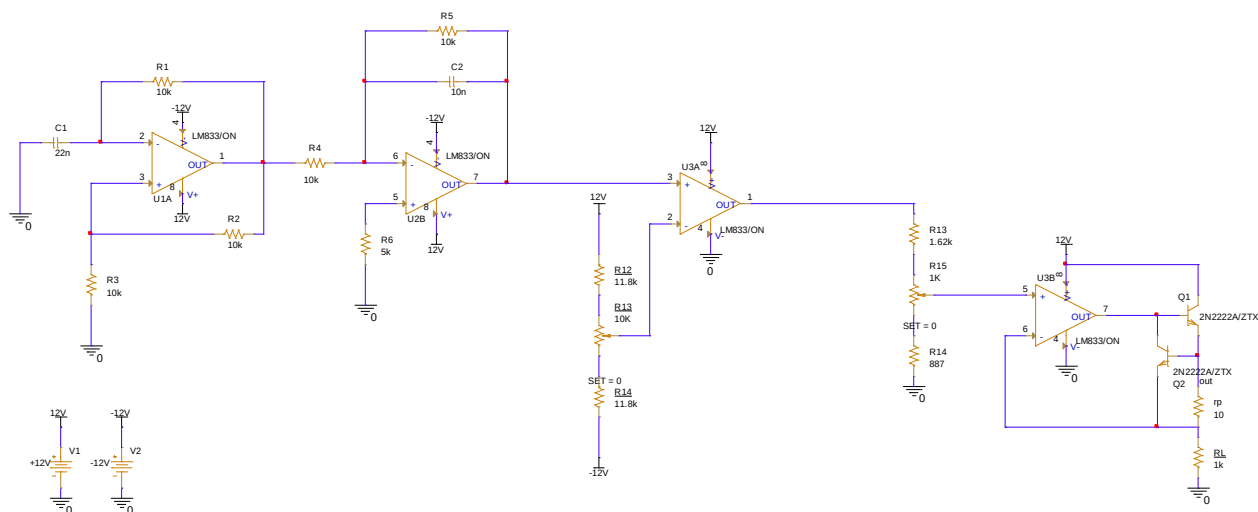


Figure 1. PWM Generator

4. Design calculations and component justification

4.1 Square wave generator

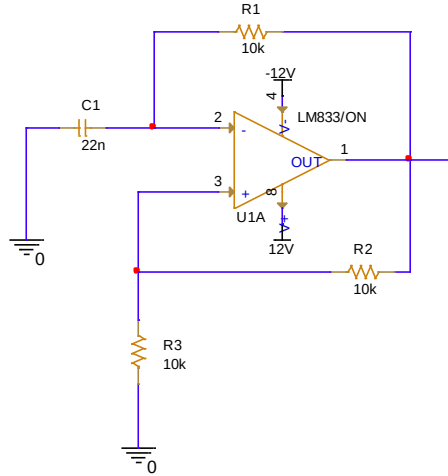


Figure 2 Square wave generator

In figure 2 is designed a square wave oscillator using an operational amplifier. I chose the LM833 op-amp for its fast response and low noise characteristics, making it well-suited for signal generation applications. For the power supply, I decided to use $[-12V; +12V]$, which allows the output to swing fully between positive and negative saturation levels, ensuring a clear and symmetrical square wave. I used R2 and R3, both 10k from E96 and the frequency is 2000Hz. Therefore, the calculated value for the capacitor C1 is:

$$f = \frac{1}{2 * R1 * C1 * \ln(\frac{1+\beta}{1-\beta})}, \text{ where } \beta = \frac{R3}{R2+R3} \Rightarrow \beta = 0.5 \Rightarrow \ln(\frac{1+\beta}{1-\beta}) = \ln(3)$$

$$2 * 10^3 = \frac{1}{2 * R1 * C1 * \ln(3)} \Rightarrow 2 * R1 * C1 * \ln(3) = \frac{1}{2 * 10^3} \Rightarrow R1 * C1 = \frac{1}{2 * 2 * 10^3 * \ln(3)}$$

$$\text{I chose } R1=10k \text{ from E96} \Rightarrow C1 = \frac{1}{2 * 2 * 10^3 * 10 * \ln(3)}, \text{ where } \ln(3) \approx 1.0986$$

$$\Rightarrow C1 = 2,2756 * 10^{-5} F = 2,2756 * 10^{-5} F * 10^9 nF = 2.2756 * 10^4 nF$$

So C1=22nF from E12. In OrCad I obtained the variation for the output $V_{out} \in [-12.250; 11.751]$

4.2 Integrator

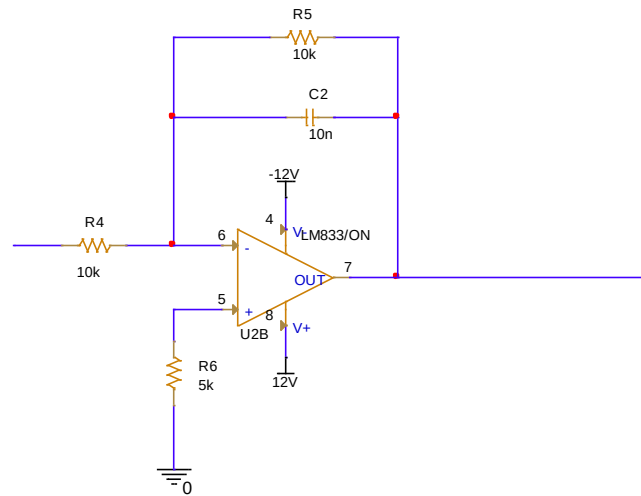


Figure 3 Integrator

In figure 3 is the second part of the PWM generator. After generating a square wave in the first stage, I designed an integrator using another op-amp from the LM833 to produce a triangular waveform. The square wave applied to the inverting input causes the capacitor C2 to charge and discharge linearly. Therefore, a triangular waveform appears at the op-amp's output.

$T = 500\mu s \Rightarrow$ the integrator's response is too slow, so I chose $T = 100\mu s$

I chose $R4 = 10k\Omega \Rightarrow T = R4 * C2 \Rightarrow 100 = 10 * C2 \Rightarrow C2 = 10nF$

$$R4 = R5 = 10k\Omega$$

$$\text{For } R6: R6 = R4 \parallel R5 = \frac{R4 * R5}{R4 + R5} = \frac{10 * 10}{10 + 10} \Rightarrow R6 = 5k\Omega$$

4.3 Comparator

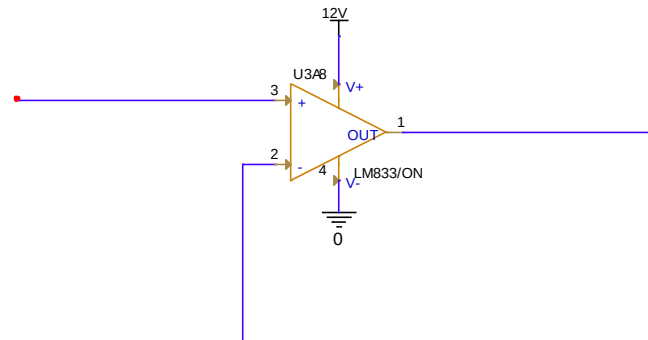


Figure4 Comparator

To complete the PWM generator, I used a comparator also implemented with an operational amplifier from the LM833 circuit. The inverting input (–) is connected to a reference voltage V_{ref} , which is derived from a voltage divider. The non-inverting input (+) is fed with the triangular waveform generated earlier by the integrator. As V_{ref} increases, the time during which the PWM signal remains high decreases, resulting in a lower duty cycle. Conversely, a lower V_{ref} leads to a higher duty cycle.

I supplied the comparator with 0 V and +12 V because the output signal later needs to be adjusted to an amplitude of [0;4 V] and [0;5 V].

4.4 Voltage divider used to adjust the PWM duty cycle

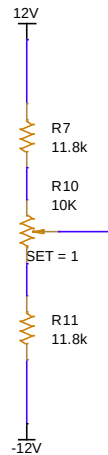
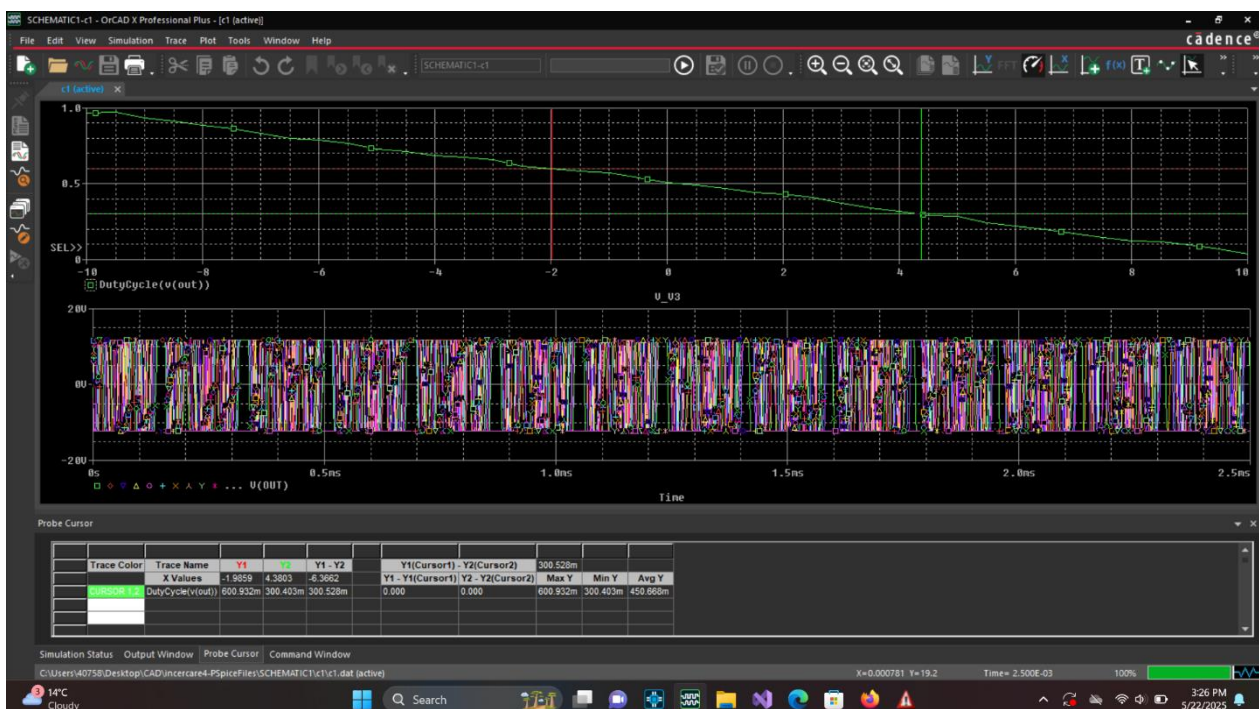


Figure 5 Voltage divider used to adjust the PWM duty cycle

An adjustable reference voltage ranging from -12 V to $+12\text{ V}$ is used by the comparator to regulate the duty cycle of the PWM signal. To find the reference voltages needed for achieving 30% and 60% duty cycles, I applied a parametric simulation approach. Instead of using a fixed voltage divider, I connected a variable DC voltage source to the inverting input of the comparator. The simulation was configured in OrCAD using the DC Sweep function to vary the input voltage. For measuring the duty cycle, I used the Performance Analysis tool and added a “Duty Cycle” measurement for the comparator’s output. By observing the simulation results with the cursor, I accurately identified the reference voltage values that correspond to duty cycles of 30% and 60%.



So, for a duty cycle of 30% $V_{ref} = 5.63V$ and for a duty cycle of 60%, $V_{ref} = -2.9V$

The values that I used are - for potentiometer $R_{10} = 10k\Omega$.

-for $R_{11} = 11.8k\Omega$

For R_7 the formula is: $V_{ref} = \frac{R_{11} * \alpha + R_{10}}{R_7 + R_{11} + R_{10}} * (V_-) + \frac{R_7 + R_{11} * (1 - \alpha)}{R_7 + R_{10} + R_{11}} * (V_+)$

$$-2.9 = \frac{10 * 10^3 * 1 + 11800}{R_7 + 10 * 10^3 + 11800} * (-12) + \frac{R_7}{R_7 + 11800 + 10 * 10^3} * 12$$

$$\Rightarrow R_7 = 13,3k\Omega$$

But with this value, the duty cycle wasn't 30% and 60%, so I chose $R_7 = 11.8k\Omega$. This value wasn't calculated using a formula but was obtained through practical calibration to achieve the desired duty cycle.

4.5 Voltage divider used to adjust signal amplitude

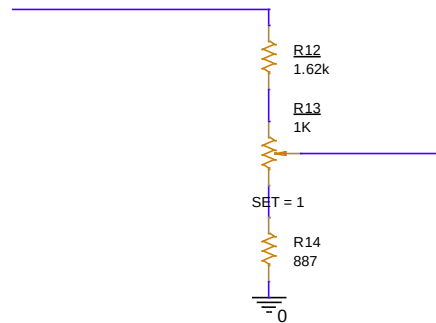


Figure 6 Voltage divider used to adjust signal amplitude

To adjust the amplitude of the generated PWM signal, which ranges from 0 to 12V, a voltage divider was implemented using resistors R12, R13 and R14. This configuration allows the use of a potentiometer at the point marked "SET," which acts as an adjustable wiper between R12 and R14. As a result, the output signal amplitude can be smoothly varied. Depending on the wiper position, the signal amplitude can be reduced linearly to a maximum of either 4V or 5V.

I chose $R_{14} = 1k\Omega$ and $R_{13} = 1k\Omega$ (standard values from E24 series)

I calculated R_{12} using the formula: $V_{out} = V_{in} * \frac{R_{14} + R_{13} * (1 - \alpha)}{R_{12} + R_{13} + R_{14}}$

For $V_{out} = 4V$, $\alpha = 1 \Rightarrow$

$$4V = 12V * \frac{1k\Omega}{2k\Omega + R_{12}} \Rightarrow 2 + R_{12} = \frac{12}{4}$$

$$\Rightarrow R_{12} = 3 - 2 = 1k\Omega$$

For $V_{out} = 5V, \alpha = 0 \Rightarrow$

$$5 = 12V * \frac{1k\Omega + 1k\Omega}{2k\Omega + R_{12}} \Rightarrow 2 + R_{12} = \frac{24}{5} \Rightarrow 2 + R_{12} = 4,8k\Omega \Rightarrow R_{12} = 2.2k\Omega$$

I chose a standard value from E24 between $1k\Omega$ and $2.2k\Omega$ which is $1.62k\Omega$, then I chose two standard values from the E96 for R_{13} and R_{14} suitable for adjusting the signal amplitude to $[0;4V]$ and $[0;5V]$.

4.6 Protection circuit

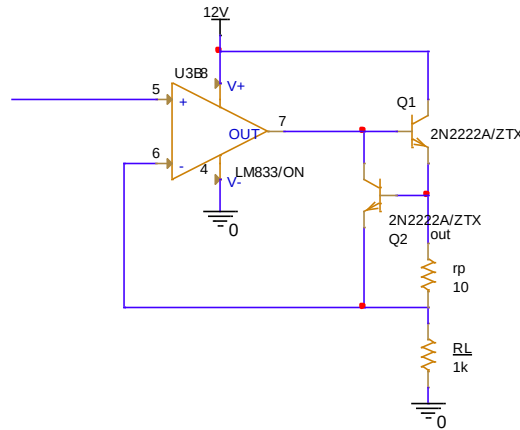


Figure 7 Protection circuit

To allow safe connection of a variable R_L (between 100Ω and $1k\Omega$) without affecting the circuit's operation, an active protection stage was implemented. It consists of an operational amplifier and two NPN transistors (2N2222), configured to limit the current through the load.

The resistor r_p , placed between the emitter of the lower transistor and ground, enables current sensing. If the current becomes too high, the voltage across r_p increases, triggering feedback to the op-amp, which reduces the drive to the transistors, thereby protecting both the load and the power supply.

$$R_L = \begin{cases} 100\Omega \\ 1k\Omega \end{cases}$$

$$V_{out} = \begin{cases} 4V \\ 5V \end{cases}$$

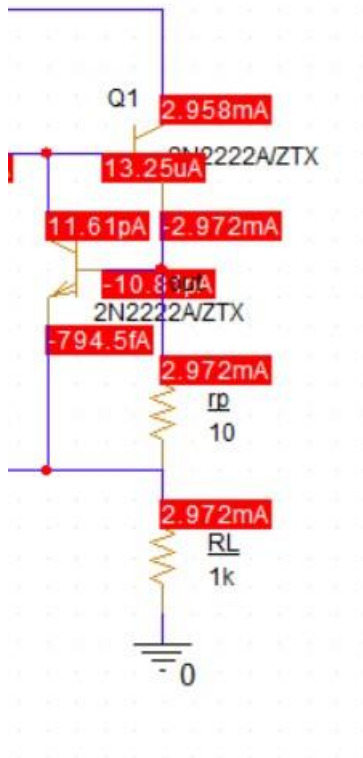
$$\Rightarrow I_{omax} = \frac{5V}{100\Omega} = 50mA$$

$$I_{omax} = \frac{0.7V}{r_p} \Rightarrow r_p = \frac{0.7V}{0.006A} = 11.6\Omega$$

$\Rightarrow r_p = 10\Omega$ (standard value from E24)

A Bias Point analysis was conducted in OrCAD to assess the protection circuit's behavior under short-circuit conditions. For this purpose, the load resistor R_L was substituted with a direct ground connection, effectively simulating a no-load situation.

After completing the simulation, the "Enable Bias Current Display" feature was activated, allowing the bias current values to be visualized directly on the schematic.



The current at the emitter of the Q2N2222 transistor is 2.59 mA, which is lower than the calculated value which is 80mA. Therefore, the circuit operates correctly.

5. Simulations

5.1 Time Domain simulations

5.1.1 Voltage variation after rectangular signal generator

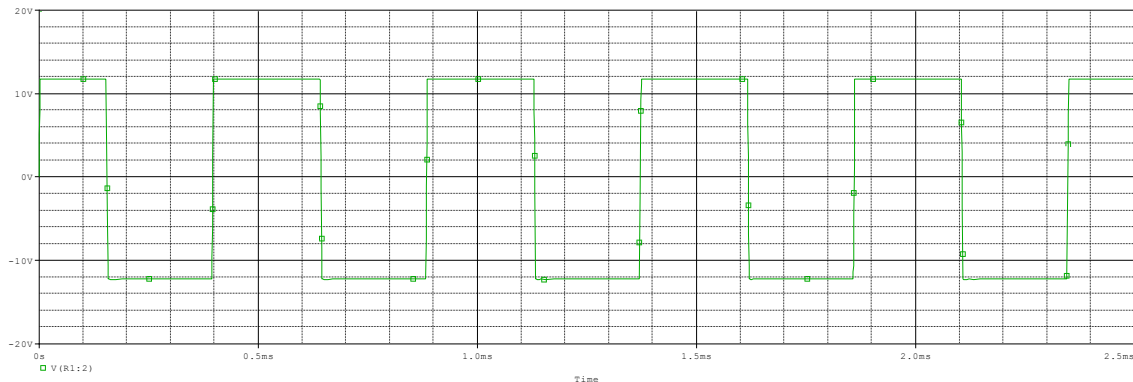


Figure 8 Square wave signal

The simulation was performed in the Time domain, with a run to time set to 2.5ms. The generated PWM signal has a frequency of 2000 Hz, corresponding to a period of 0.5ms. The waveform is rectangular, as expected for a PWM signal, and it varies between approximately $[-12.2V; 11.7V]$.

These are the simulation settings:

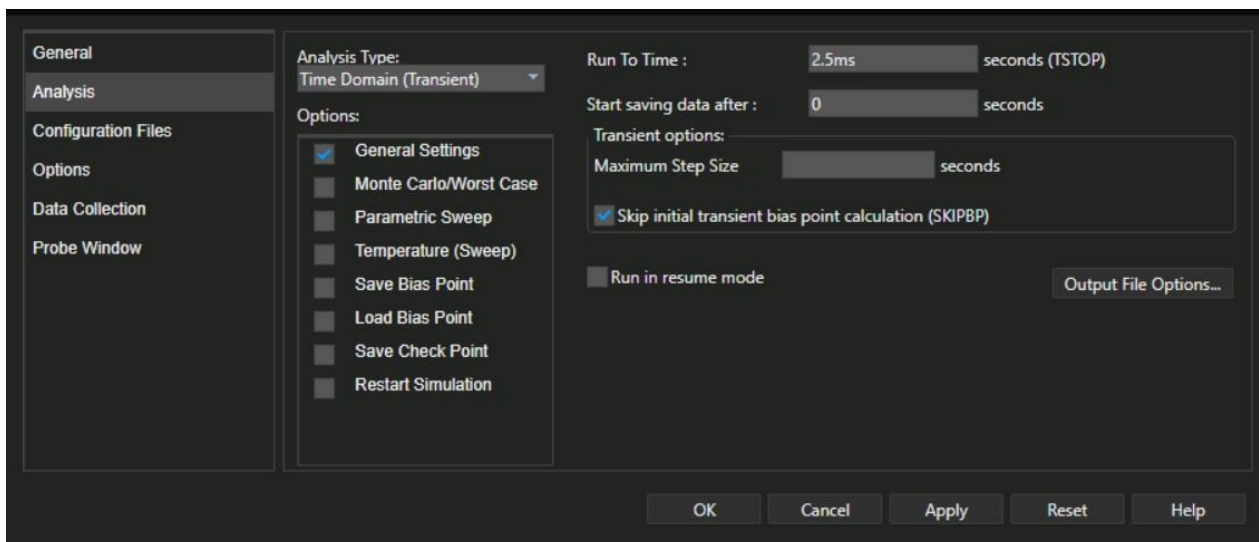


Figure 9 Simulation profile

5.1.2 Voltage variation after integrator

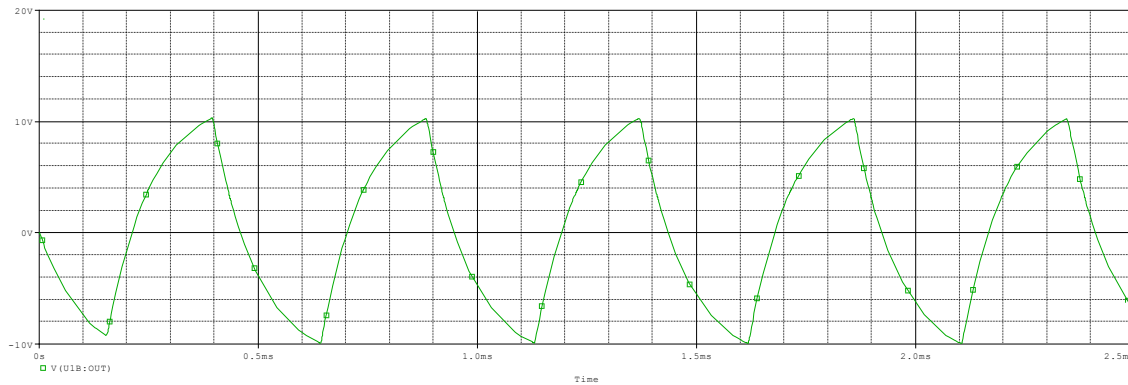


Figure 11 Triangular signal

In this simulation, a triangular waveform was generated using an integrator circuit. The simulation was performed in the Time domain, and the resulting waveform oscillates between approximately $[-9.907 \text{ V}; 10.317 \text{ V}]$. The signal exhibits a smooth linear rise and fall, which is characteristic of triangular waveforms, confirming the correct operation of the integrator stage.

5.1.3 Voltage variation after voltage divider for modifying duty cycle

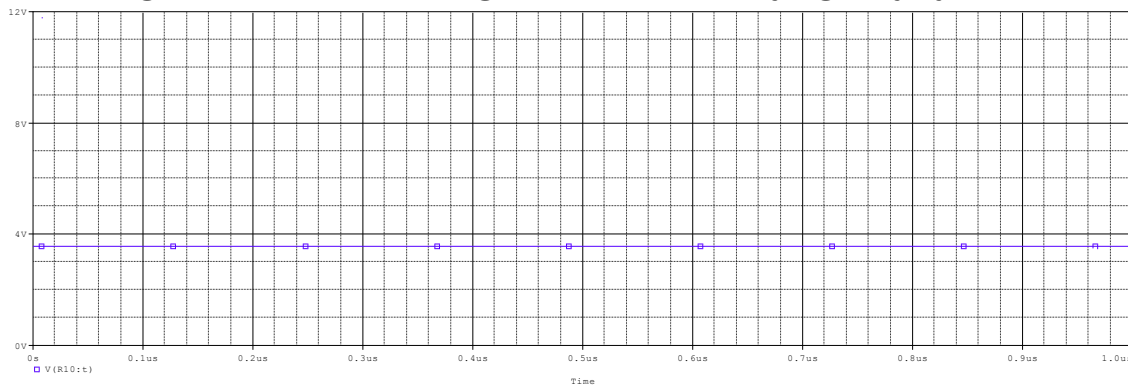


Figure 12 Voltage after voltage divider with $\alpha = 0$

In Figure 12, the voltage V_{ref} is shown for $\alpha = 0$. This voltage corresponds to a duty cycle of 30%.

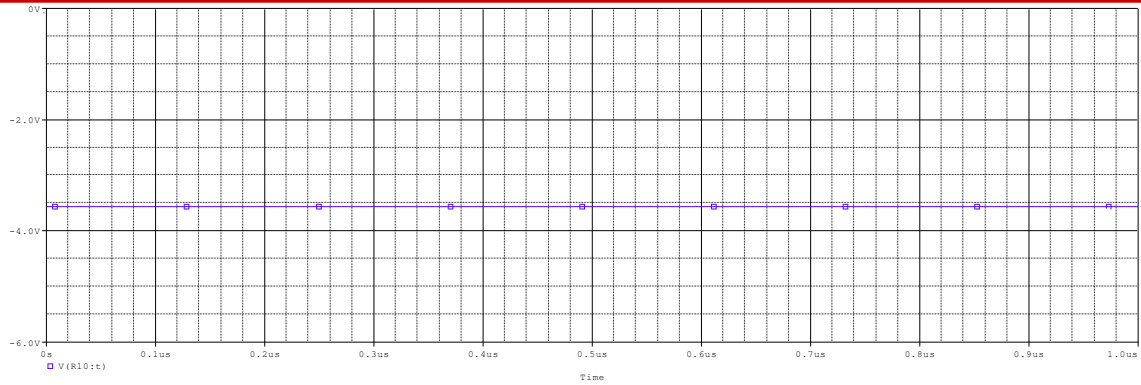


Figure 11 DC voltage after voltage divider with $\alpha = 1$

In Figure 11, the voltage V_{ref} is shown for $\alpha = 0$. This voltage corresponds to a duty cycle of 60%.

5.1.4 Voltage variation after comparator

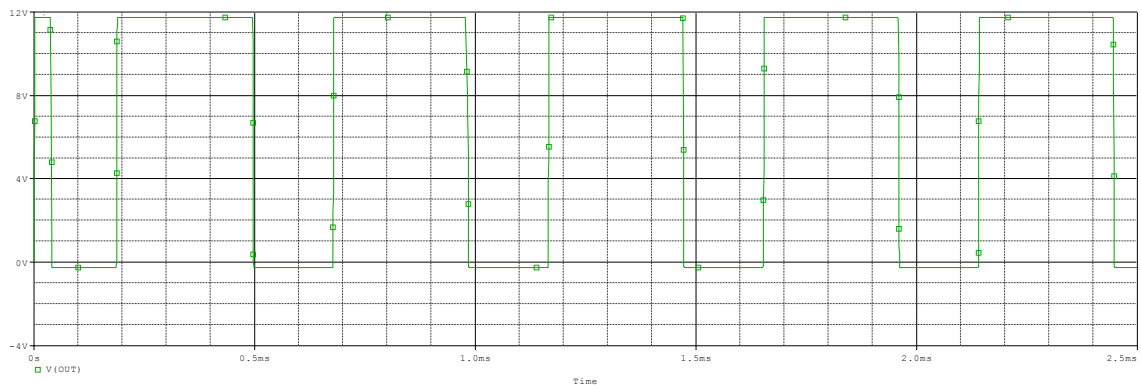


Figure 12 Voltage after comparator when $\alpha = 1$

This simulation illustrates the behavior of the comparator when the $10k\Omega$ potentiometer is set to the "SET = 1" position. A PWM signal is generated, its shape is directly influenced by the potentiometer adjustment.

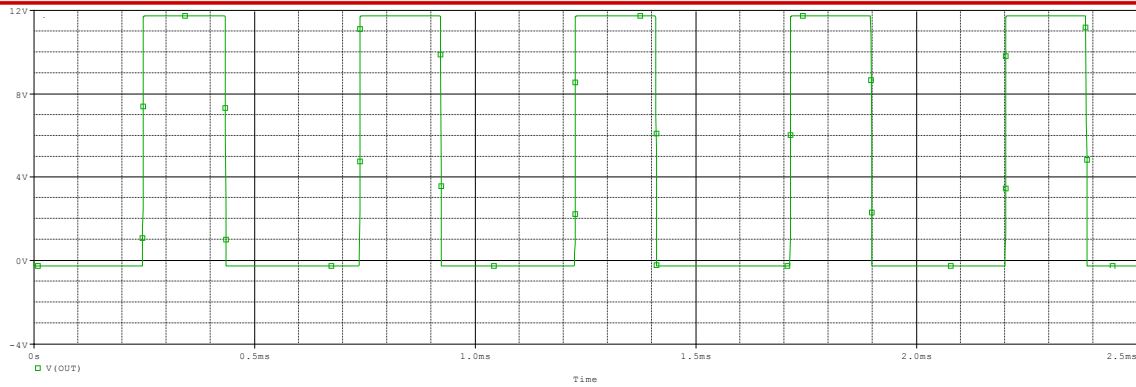


Figure 13 Voltage after comparator when $\alpha = 0$

This simulation shows the output of the comparator when the potentiometer is set to the "SET = 0" position. As a result of this setting, the reference voltage at the comparator input is significantly lower, which causes the output PWM signal to have a shorter high-state duration.

5.1.5 Voltage after voltage divider for adjusting the amplitude

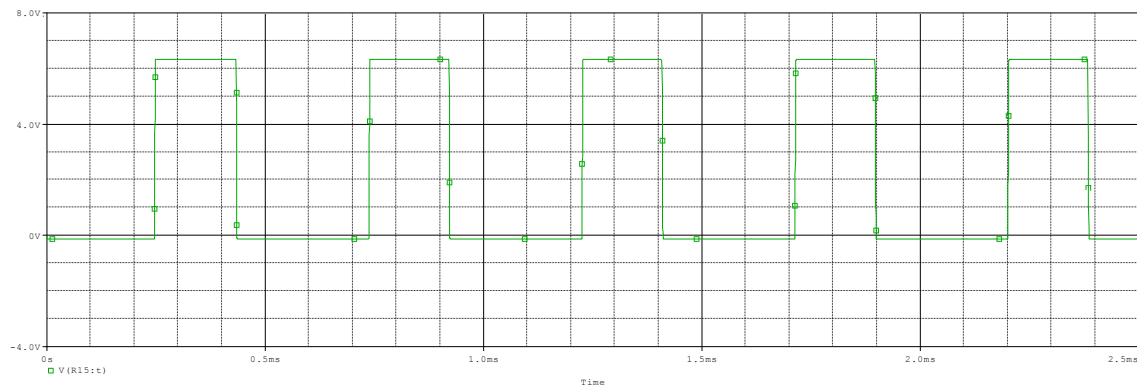


Figure 14 Voltage after voltage divider when $\alpha = 0$

This simulation shows the output of the comparator using a 1k Ω potentiometer with the wiper set to position 0. The goal was to limit the PWM signal amplitude to a range between [0;5V]. Multiple calculations and tests were performed using standard resistor values and voltage divider formulas, but none of the combinations provided the exact desired amplitude. As a compromise, the resistor values $R13 = 1.62\text{k}\Omega$ and $R14 = 887\Omega$ were selected, as they yielded the closest approximation to the target amplitude. This choice ensured a functional and stable signal within an acceptable voltage range for the application.

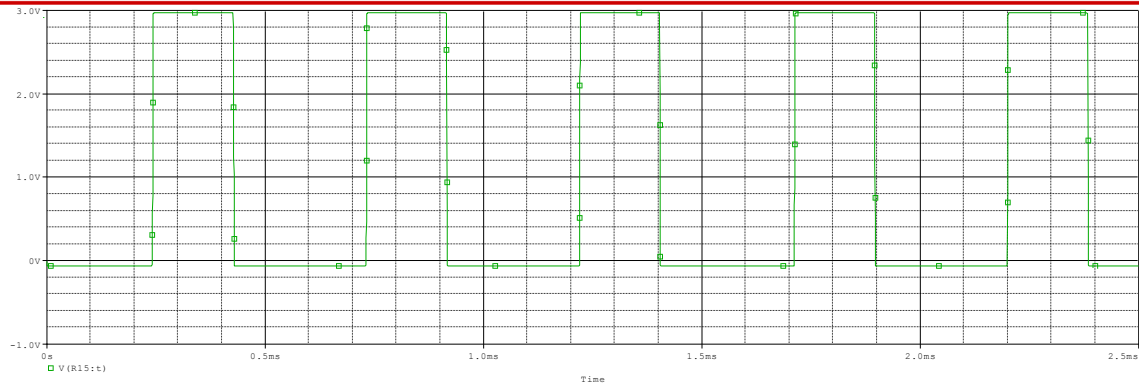


Figure 15 Voltage after voltage divider when $\alpha = 1$

In figure 15 is the output of the PWM signal when the 1k Ω potentiometer is set to position 1. In this configuration, the voltage divider using resistors R12 = 1.62k Ω and R14 = 887 Ω produces an output amplitude in the desired range of [0V;4V].

5.2 Monte Carlo simulations

5.2.1 Monte Carlo simulations for duty cycle

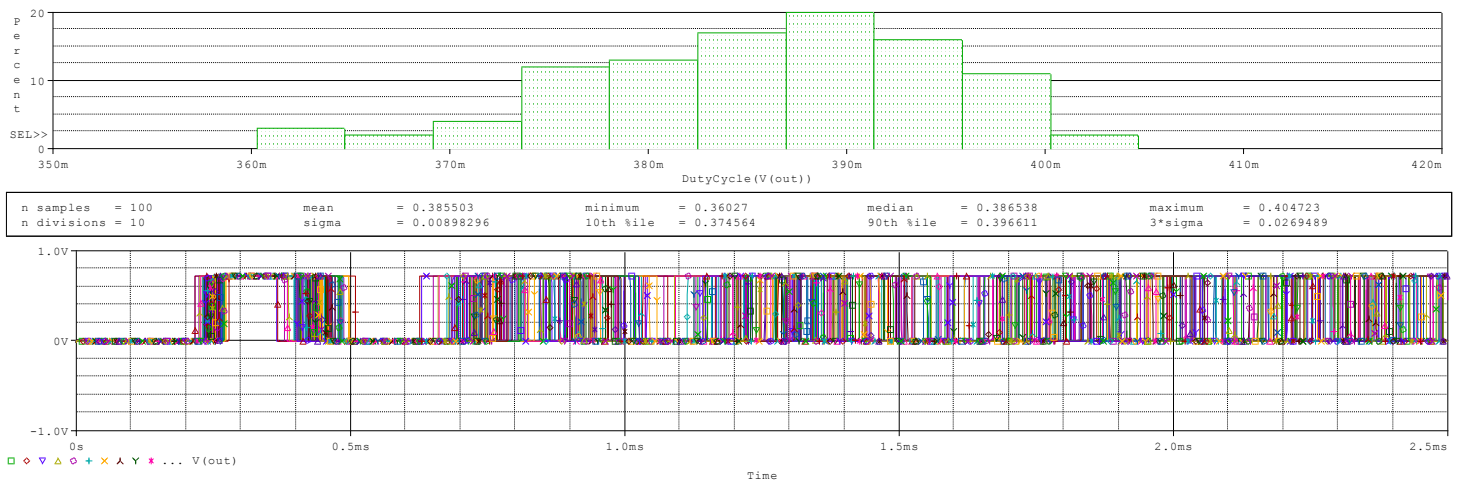


Figure 16 Monte Carlo for duty cycle of 30%

To assess the influence of component tolerances on the stability of the PWM signal, a Monte Carlo simulation with 100 iterations was carried out in OrCAD. The analysis focused on the signal at the out node. Despite the presence of component tolerances, the PWM signal exhibited only minor deviations in duty cycle, consistently remaining close to the target value of 30%.

The key statistical outcomes derived from the simulation are summarized below.

Average duty cycle: 38.6%

Minimum: 36%

Maximum: 40%

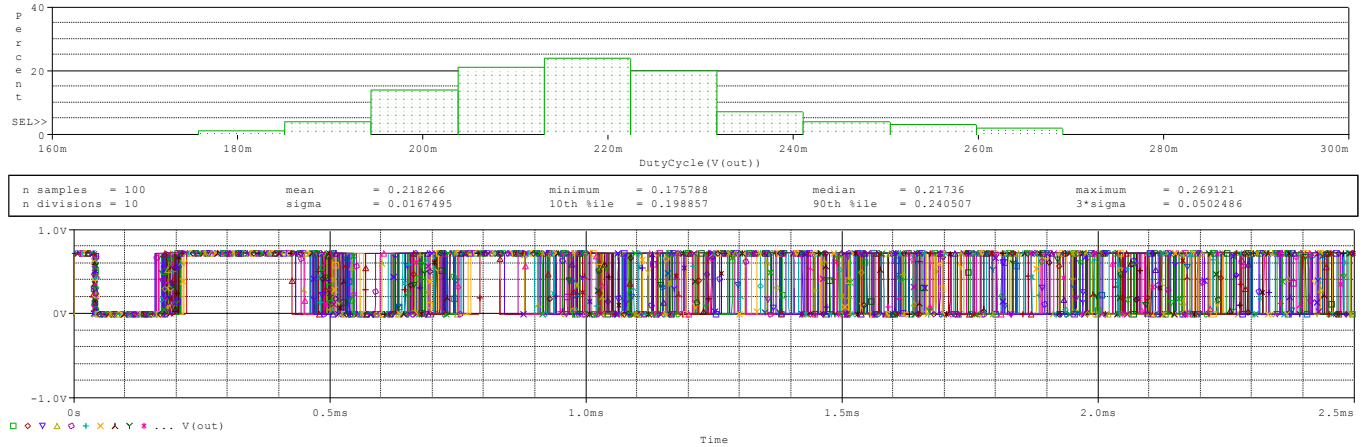


Figure17 Monte Carlo for duty cycle of 60%

Average duty cycle: 21%

Minimum: 17%

Maximum: 27%

5.2.2 Monte Carlo simulation for amplitude

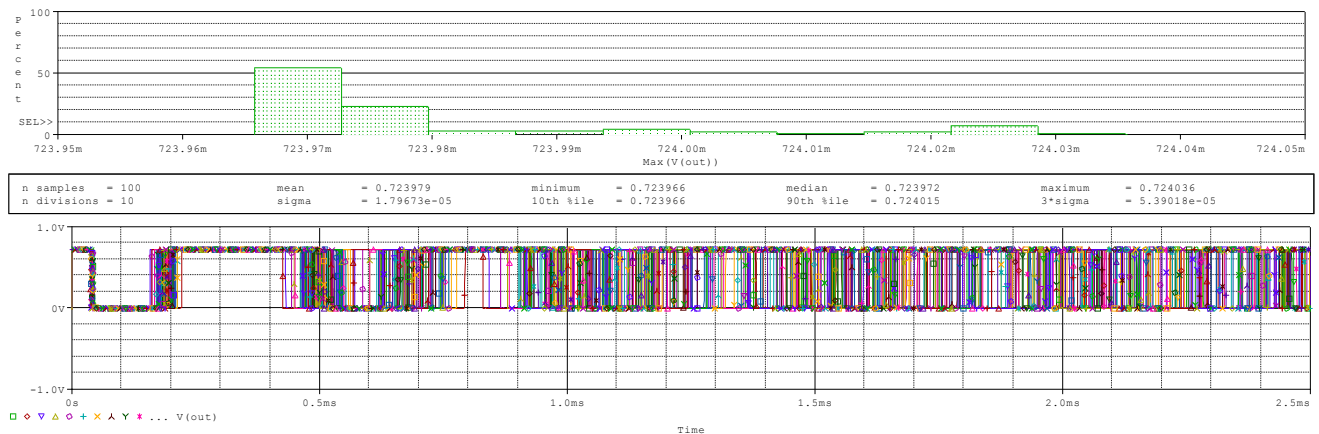


Figure 2 Monte Carlo for amplitude of 4V

Average amplitude: 72.3V

Minimum: 72.3V

Maximum: 72.4V

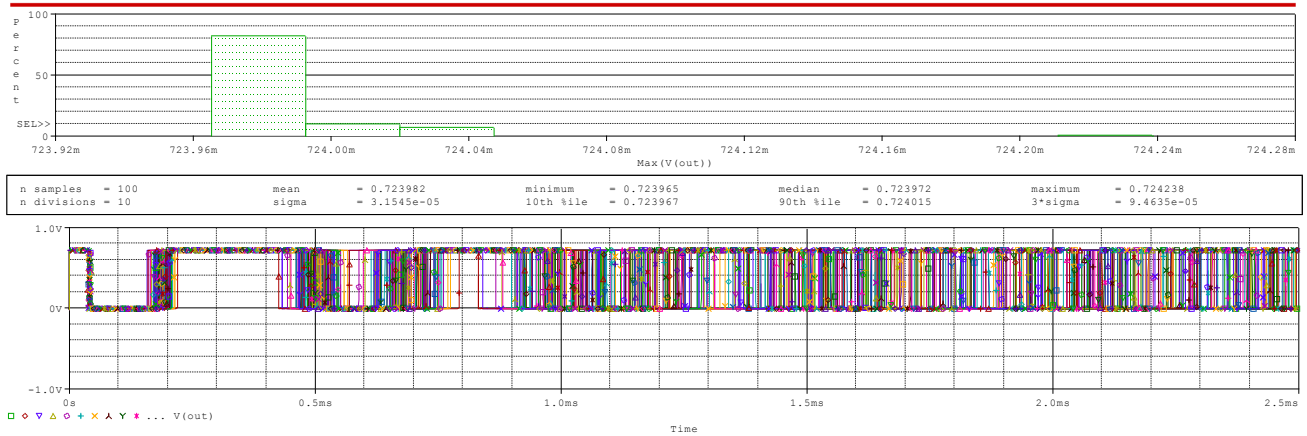


Figure 3 Monte Carlo for amplitude of 5V

Average amplitude: 72.3V

Minimum: 72.3V

Maximum: 72.4V

5.2.3 Monte Carlo simulation for period

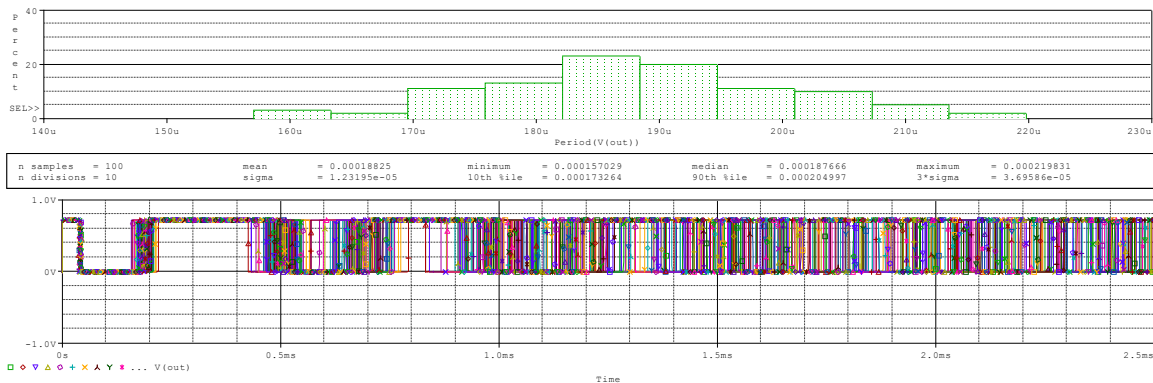


Figure 19 Monte Carlo for period

Average period: 100.86μs

Minimum: 100.57μs

Maximum: 200.19μs

6. Bill of Materials

Index	Cantitate	Descriere	Preț unitar	Preț total RON
1	1	CAP CER 0805 10NF 50V COG 1%	6.15	6,15 lei
2	5	RES 10K OHM 5% 1/4W AXIAL	0.43	2,15 lei
3	1	RES 5K OHM 1/8W 1% AXIAL	2.6	2,60 lei
4	2	RES SMD 11.8K OHM 1% 1/8W 0805	0.43	0,86 lei
5	1	RES 887 OHM 1% 1/4W AXIAL	0.43	0,43 lei
6	1	RES 10 OHM 1% 1/4W AXIAL	0.43	0,43 lei
7	1	RES 1K OHM 5% 1/4W AXIAL	0.43	0,43 lei
8	4	IC AUDIO 2 CIRCUIT 8SOIC	5.24	20,96 lei
9	2	IC AUDIO 2 CIRCUIT 8SOIC	5.24	10,48 lei
10	1	RES 10 OHM 5% 1/4W AXIAL	0.43	0,43 lei
11	1	FILM CAP MKP BOX 22NF 5% 1000VDC	3.2	3,20 lei

7. Conclusion

The analysis of the PWM generator, performed both in the time domain and through Monte Carlo simulations, demonstrated the circuit's stable performance under component tolerance variations.

Two duty cycle values were tested 30% and 60% along with output signal amplitudes of 4 V and 5 V. The results showed consistent and reliable PWM signal generation, with only minor variations around the target values, confirming the robustness of the circuit design and its resilience to component tolerances.

Notably, the signal period remained stable around the nominal value of 100 μ s. This highlights the circuit's high stability and makes it well-suited for applications requiring precise and reproducible control signals.

8. Bibliography

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<https://www.youtube.com/watch?si=QqBeIyhzTyWbdeu6&v=Zep3ikgHutU&feature=youtu.be>

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<https://chatgpt.com/>